More Contradiction; some Induction

WTP P

Assume $\neg P$

Argue

Conclude Q

Observe Q is false

Conclude P

$$(\neg P \Rightarrow Q) \wedge \neg Q \Rightarrow P$$

$P \Rightarrow Q$ $\qquad$ $\neg Q \Rightarrow \neg P$

$\cancel{\neg P \Rightarrow \neg Q}$

Let $n \in \mathbb{Z}$.

Prove that n and $n+1$ have no common prime factors.

Assume n and $n+1$ have a common prime factor p. Then $p \mid n$ and $p \mid n+1$. $\Rightarrow p \mid (n+1)-n$ $\Rightarrow p \mid 1$

By the 2 out of 3 rule, $p \mid 1$.

But the smallest prime is $p = 2$, so this is a contradiction.

$\boxtimes$

(10) Let a, b, c be integers satisfying $a^2 + b^2 = c^2$. Show that abc must be even. (Harder problem: show that a or b must be even.)

Proceed by contradiction. Assume a, b, and c or all odd, then a^2, b^2 r c^2 are odd. But since $a^2+b^2=c^2$, the LHS is even, and the RHS is odd. This is a contradiction. $\blacksquare$

$\neg(P \Rightarrow Q) = P \wedge \neg Q$

(9) Prove that if $r^3 + r + 1 = 0$ then r is irrational.

Proceed by contradiction.

Assume $r^3 + r + 1 = 0$ and $r \in \mathbb{Q}$.

Since $r \in \mathbb{Q}$, $\exists a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $r = a/b$. Moreover, WMA (we may assume) that a and b are "reduced", i.e., they have no common factors. (In particular, at least one of a or b is odd.) Thus, $\left(\frac{a}{b}\right)^3 + \frac{a}{b} + 1 = 0$.

Clearing denominators

$$a^3 + ab^2 + b^3 \overset{(*)}{=} 0$$

Case 1: a is even. Then b is odd. Then the LHS of $(*)$ is odd, but 0 is even. This is a contradiction.

Case 2: b is even. This is the same as case 1.

Case 3: a and b are both odd. But then the LHS of $(*)$ is odd, but the RHS is even, a contradiction.

We conclude that $r \notin \mathbb{Q}$. ∎

$\neg P \Rightarrow (Q_1 \vee Q_2 \vee Q_3)$

$\neg(Q_1 \vee Q_2 \vee Q_3) = \neg Q_1 \wedge \neg Q_2 \wedge \neg Q_3$

$a^{\log_a b} = b$

$\log_a x = b \iff x = a^b$ $\qquad$ $\log_a (a^b) = b$

a_1 $\qquad$ $\log_{10} 100 = 2$ $\qquad$ a

$\log_a x$ and a^x are inverse

(13) Prove that $\log_{10} 7$ is irrational.

Proceed by contradiction. Assume that $\log_{10} 7 \in \mathbb{Q}$. Thus, $\exists\, a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $\log_{10} 7 \overset{(*)}{=} a/b$. WMA a & b have no common prime factors. Moreover, at least one of a or b is not negative.

Applying 10^x to equation $(*)$ gives

$$7 = 10^{\log_{10} 7} = 10^{a/b}. \text{ Thus } 7^b \overset{(*)}{=} 10^a.$$

① If a or b is negative, we get that an integer is equal to a non integer, ↯

② If a and b are both not negative, then, since $b > 0$, the LHS of $(*)$ is a multiple by 7, but the RHS is not a multiple of 7, ↯.

We conclude that $\log_{10} 7 \notin \mathbb{Q}$. ∎

Induction $\forall n \in \mathbb{Z}_{>0}$

$P(n)$

$$1+2+3+\dots+n = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

Base case

$$1 = \frac{1(1+1)}{2} = \frac{1\cdot 2}{2} = 1$$

$$1+2 = \frac{1\cdot 2}{2} + 2 = 2\left(\frac{1}{2}+1\right) = 2\left(\frac{1+2}{2}\right) = \frac{2\cdot 3}{2}$$

$$1+2+3 = \frac{2\cdot 3}{2} + 3 = 3\left(\frac{2}{2}+1\right) = 3\left(\frac{2+2}{2}\right) = \frac{3\cdot 4}{2}$$

$$1+2+3+4 = \frac{3\cdot 4}{2} + 4 = 4\left(\frac{3}{2}+1\right) = 4\left(\frac{3+2}{2}\right) = \frac{4\cdot 5}{2}.$$

Proof that $P(n-1) \Rightarrow P(n)$

Proof: Suppose we already knew that

Assum $P(n-1)$

$$1+2+\dots+(n-1) = \frac{(n-1)n}{2}.$$

Adding n to both sides gives

$$1+2+\dots+(n-1)+n = \frac{(n-1)n}{2} + n,$$

Proof

The RHS of this is

$$n\left(\frac{n-1}{2}+1\right) = n\left(\frac{n-1+2}{2}\right) = \frac{n(n+1)}{2}.$$

We conclude that $\forall n \in \mathbb{Z}_{>0}$,

$$1+2+\dots+n = \frac{n(n+1)}{2} \quad \blacksquare \quad P(n)$$

$P(n) = \cancel{\frac{n(n+1)}{2}}$

" $1+2+\dots+n = \frac{n(n+1)}{2}$ "

Define a sequence $a_1, a_2, a_3, \ldots$ as

$$\boxed{\begin{aligned} a_1 &= 2 = 2^1 \\ a_n &= 2 \cdot a_{n-1} \end{aligned}}$$

"recursive definition"

$a_2 = 2 \cdot a_1 = 2 \cdot 2 = 4 = 2^2$

$a_3 = 2 \cdot a_2 = 2 \cdot 4 = 8 = 2^3$

$a_4 = 2 \cdot a_3 = 2 \cdot 8 = 16 = 2^4$

$\vdots$

Claim: $\forall n \in \mathbb{Z}_{>0}, \ a_n = 2^n$

Idea: $2^{n-1} \cdot 2 = 2^n$

$P(n) =$ "$a_n = 2^n$"

$\forall n \in \mathbb{Z}_{>0}, P(n)$

BASE CASE

$P(1) =$ "$a_1 = 2^1$" ← true by definition

Assume that $P(n)$ is true, i.e., assume that $a_n = 2^n$. Then $a_{n+1} = 2 \cdot a_n$.

By our hypothesis, $a_{n+1} = 2 \cdot a_n = 2 \cdot 2^n = 2^{n+1}$.

Thus $P(n+1)$ is also true

We conclude that $\forall n > 0, P(n)$.

$P(n) \Rightarrow P(n+1)$

$\boxed{\text{Induction}}$ 4

Let $P(n)$ be a statement which depends on an integer n

$\boxed{\text{Ex}}$ $P(n) =$ "$a_n = 2^n$" or

$P(n) =$ "$1+2+\cdots+n = \frac{n(n+1)}{2}$"

$\boxed{\text{Goal}}$ Prove $P(n)$ $\forall n \in \mathbb{Z}_{>0}$

Step 1: Prove $P(1)$ "Base case"

Step 2: Prove "$P(n-1) \Rightarrow P(n)$" "Inductive step"

"Induction" = $P(1)$ and "$P(n-1) \Rightarrow P(n)$" $\Rightarrow$

$\forall n \in \mathbb{Z}_{>0}, P(n)$

$P(1), P(2), P(3), P(4), P(5), \ldots, P(n-1), P(n), P(n+1), \ldots$

$P(a-1) \Rightarrow P(a)$

$P(a) \Rightarrow P(a+1)$

WTP $\forall n \in \mathbb{Z}_{>1}, P(n)$

$P(2) \wedge P(n) \Rightarrow P(n+1)$

WTP $\forall$ even $n \in \mathbb{Z}_{>0}$, $P(n)$.

$P(2) \wedge P(n) \Rightarrow P(n+2)$

3. We want to prove, by induction, that, for every positive integer n,

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

a) What is the open statement "$P(n)$"?

P(n) =

b) What is the statement "$P(1)$"? Why is $P(1)$ true?

P(1) =

c) What is the inductive step? Write out your assumption, your desired conclusion, and the inductive step (i.e., the proof that $P(n-1) \Rightarrow P(n)$).

Assume that

We want to show that

(Inductive step)